EXPERIMENT 9

Contiguous Memory Allocation: FIRST FIT

AIM:

To implement the First Fit memory allocation algorithm in C

PROCEDURE:

1. Start Fedora Linux using Oracle VirtualBox.
2. Open the terminal and create a C program.
3. Enter the number of memory blocks and processes.
4. Read the memory block sizes and process sizes.
5. Check each memory block sequentially for every process.
6. Allocate the process to the first block having sufficient space.
7. Update the remaining size of the allocated block.
8. Display the memory allocation details.
9. Stop.

PROGRAM:

```
#include <stdio.h>

int main()
{
    int b[20], p[20], a[20];
    int nb, np, i, j;

    printf("Enter number of blocks: ");
    scanf("%d", &nb);

    printf("Enter block sizes:\n");
    for(i = 0; i < nb; i++)
        scanf("%d", &b[i]);

    printf("Enter number of processes: ");
    scanf("%d", &np);

    printf("Enter process sizes:\n");
    for(i = 0; i < np; i++)
        scanf("%d", &p[i]);

    for(i = 0; i < np; i++)
    {
        a[i] = -1;

        for(j = 0; j < nb; j++)
        {
```

```
            if(b[j] >= p[i])
            {
                a[i] = j;
                b[j] -= p[i];
                break;
            }
        }
    }

    printf("\nProcess\tSize\tBlock\n");

    for(i = 0; i < np; i++)
    {
        printf("P%d\t%d\t", i+1, p[i]);

        if(a[i] == -1)
            printf("Not Allocated\n");
        else
            printf("%d\n", a[i]+1);
    }

    return 0;
}
```

OUTPUT:



RESULT:

Thus, the First Fit memory allocation algorithm was implemented and executed successfully using C in Fedora Linux.